Rethinking Pose Refinement in 3D Gaussian Splatting under Pose Prior and Geometric Uncertainty

Mangyu Kong[1] Jeawon Lee[1] Seongwon Lee[2*] Euntai Kim[1,3*]
[1]Yonsei University [2]Kookmin University [3]Korea Institution of Science and Technology

Abstract

3D Gaussian Splatting (3DGS) has recently emerged as a powerful scene representation and is increasingly used for visual localization and pose refinement. However, despite its high-quality differentiable rendering, the robustness of 3DGS-based pose refinement remains highly sensitive to both the initial camera pose and the reconstructed geometry. In this work, we take a closer look at these limitations and identify two major sources of uncertainty: (i) pose prior uncertainty, which often arises from regression or retrieval models that output a single deterministic estimate, and (ii) geometric uncertainty, caused by imperfections in the 3DGS reconstruction that propagate errors into PnP solvers. Such uncertainties can distort reprojection geometry and destabilize optimization, even when the rendered appearance still looks plausible. To address these uncertainties, we introduce a relocalization framework that combines Monte Carlo pose sampling with Fisher Information–based PnP optimization. Our method explicitly accounts for both pose and geometric uncertainty and requires no retraining or additional supervision. Across diverse indoor and outdoor benchmarks, our approach consistently improves localization accuracy and significantly increases stability under pose and depth noise.

1. Introduction

Visual localization aims to estimate the 6-DoF camera pose of a query image within a known scene and plays a key role in AR/VR, autonomous driving, and robotics. A main factor governing the performance of localization methods is how the scene is represented. Traditional structure-based pipelines rely on sparse 3D point clouds reconstructed via Structure-from-Motion [41, 43]. Local features extracted from the query image are matched to reference images or 3D points [10, 39, 42], and the resulting correspondences are used to estimate the camera pose through PnP algorithms [11, 12, 24]. Scene Coordinate Regression (SCR) approaches [3–5] extend this paradigm by predicting dense scene coordinates, and Absolute Pose Regression (APR) methods [6, 8, 18, 26, 45] learn to map images directly to poses. These approaches illustrate how advances in map representation, from sparse geometric maps to implicitly learned coordinate or pose models, continue to shape the design of localization systems.

*Corresponding authors.

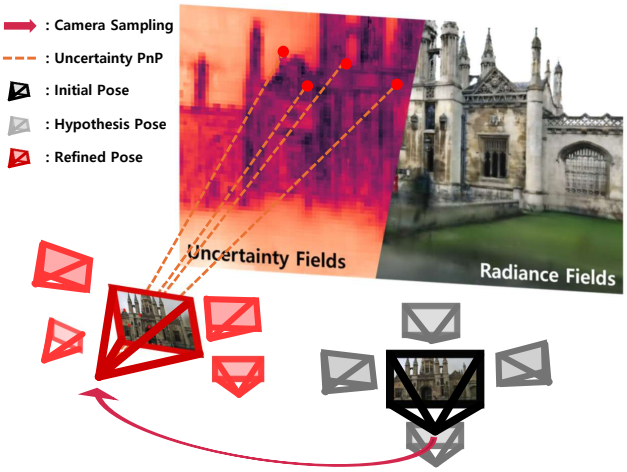

Figure 1. **UGS-Loc.** Our proposed framework refines camera pose by incorporating pose prior uncertainty via Monte Carlo sampling and geometric uncertainty via Uncertainty fields, achieving robust localization withouts retraining.

Recently, neural scene representations such as NeRF [36] and 3DGS [19] emerged as a new type of scene representation, and they triggered the development of new pose-refinement methods. These methods can be summarized as render-and-compare. They first render a synthetic view from an initial pose obtained by retrieval [1, 23], APR, or SCR, and then align this view to the query image using photometric loss [28, 50, 59] or feature distances [7, 31, 53]. More recent methods [16, 26, 30] extend this idea: they find dense 2D–2D correspondences [25, 49] between the query and the rendered view, lift these matches into 3D using rendered depth, and solve

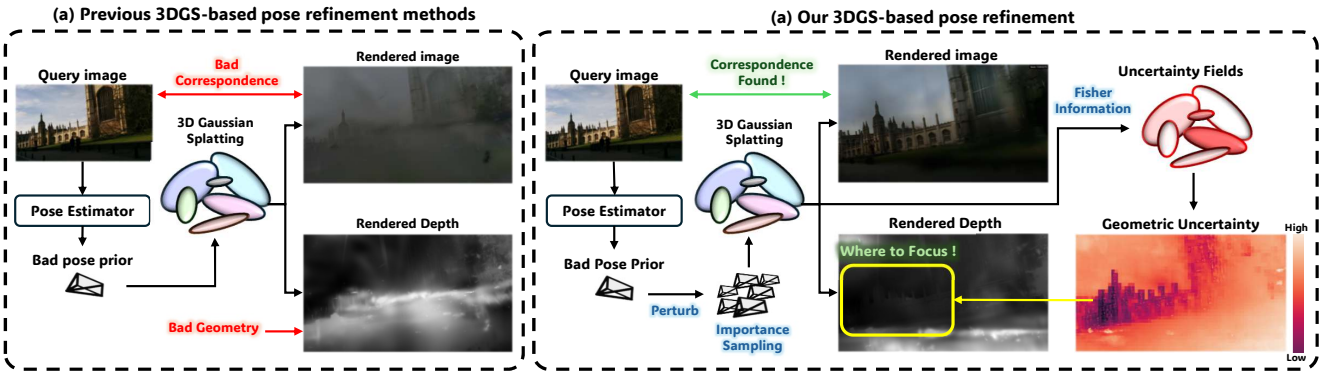

Figure 2. **Comparison between Previous 3DGS-based Pose Refinement and Our Proposed UGS-Loc.** (a) Existing pipelines refine a single deterministic pose prior using rendered depth, which makes them vulnerable to inaccurate priors and geometric errors—leading to bad correspondences and unstable refinement. (b) Our method instead applies Monte Carlo pose sampling and geometric uncertainty from Fisher Information to guide correspondences and focus refinement on reliable regions, achieving robust pose estimation even under poor pose priors and uncertain geometry.

the pose with PnP-RANSAC. This correspondence-driven version of render-and-compare now leads pose refinement and achieves state-of-the-art relocalization accuracy in rendering-based pose refinement. However, when the pose refinement methods are applied to 3DGS, we believe that the existing pose refinement methods overlook some points and may degrade some performance.

3D Gaussian Splatting models geometry with explicit elliptical primitives, but the primitives are just approximate and the depth they render is not uniformly reliable. Regions reconstructed from sparse training views [56] or contaminated by dynamic objects [20] often exhibit noticeable artifacts, leading to spatial variations in geometric accuracy. Despite these spatial variations, current refinement methods treat all rendered depths as equally trustworthy when lifting 2D–2D correspondences into 3D. As a result, they ignore the fact that reconstruction fidelity can vary significantly across different parts of the scene.

This challenge becomes even more pronounced when considering the dependence on an accurate pose prior. Since 3DGS-based refinement [16, 26, 30] operates by rendering synthetic views from a single initial estimate, the quality of correspondence hinges on how well this pose is aligned with the true scene content. When the pose prior is biased or occluded by artifacts, the rendered viewpoints may exhibit limited visibility, causing feature matchers [10, 25, 49] to struggle to establish reliable correspondences..

To address these challenges, we propose UGS-Loc, an uncertainty-aware pose refinement framework tailored to 3D Gaussian Splatting. UGS-Loc represents the pose prior as a distribution and explores it through Monte Carlo sampling, enabling the refinement pipeline to recover even when the initial pose lies far from reliable regions of the scene. In parallel, we quantify geometric uncertainty directly from the Gaussian parameters via Fisher Information [17] and integrate this signal into the PnP-RANSAC process, guiding correspondence sampling toward geometrically trustworthy regions. By jointly modeling pose and geometric uncertainties inherent to 3DGS, UGS-Loc achieves robust, training-free localization that remains stable and consistent across challenging scene conditions.

2. Related Works

Visual Localization. The task of visual localization is to determine the pose of a query image in relation to a reconstructed 3D scene. A commonly used pre-built representation of the scene is a 3D point cloud reconstructed via Structure-from-Motion (SfM) [62]. Each 3D point in this cloud is associated with a set of visual descriptors from database images, allowing the system to establish reliable 2D–3D correspondences during localization. Query image keypoints are matched between the query image and the 3D model to establish 2D–3D correspondences [10, 41, 42, 51]. Scene coordinate regression (SCR) methods [4, 5, 55] train the network to acquire 2D-3D correspondences directly. The correspondences obtained from both methods are then fed into a PnP solver to estimate the camera pose. Alternatively, absolute pose regression (APR) [18, 45] predicts the 6-DoF camera pose directly from a query image through a neural network. However, these approaches often suffer from limited pose accuracy. Our method refines the poses estimated by SCR and APR methods using 3D Gaussian Splatting (3DGS) without requiring any additional training.

Radiance Field-based Visual Localization. To overcome the sparsity and limited expressiveness of traditional 3D point cloud representations, recent works have explored dense scene representations such as radiance fields for visual localization. Leveraging the powerful novel-view synthesis capability of Neural Radiance Fields (NeRF) [36]

and Gaussian Splatting (3DGS) [19], these methods utilize photo-realistic rendering for both pose refinement and data augmentation. For instance, radiance field–based augmentation enriches APR frameworks [6, 26, 37] by generating synthetic training views across diverse viewpoints and lighting conditions, improving robustness and generalization to unseen domains.

A dominant line of research adopts the render-and-compare strategy, where the initial camera pose is iteratively refined by minimizing discrepancies between the query image and novel views rendered from the estimated pose. iNeRF [59] pioneered this paradigm by optimizing the camera pose directly via inverse rendering in NeRF, inspiring a following works [28, 50]. These approaches optimizes poses directly using photometric [2] or feature-space alignment [7] to improve localization precision. In parallel, correspondence-based pipelines have emerged as a strong alternative. CROSSFIRE [38] integrates specialized feature descriptors tailored to neural scene representations, enabling accurate 2D–3D correspondence establishment followed by PnP-RANSAC for pose estimation. This design [32, 48, 61, 63] significantly boosts localization accuracy by leveraging features that are deeply aligned with the underlying radiance field.

Beyond learning scene-specific descriptors, GS-CPR [30] adopts a simpler and more efficient strategy: using a pose prior from APR or SCR, it renders high-quality synthetic views, performs direct 2D–2D matching with the query, lifts correspondences into 3D via rendered depth, and refines the pose through PnP optimization. Similar 2D-2D correspondence-driven refinement pipelines are also employed in several recent 3DGS-based localization methods [16, 26], where a coarse pose predicted by each method serves as the initial pose. Its simplicity and efficiency make this correspondence-based refinement pipeline a widely used baseline for 3DGS-based pose refinement. However, these refinement-based approaches still heavily rely on accurate pose priors or well-reconstructed scene geometry, making them vulnerable to uncertainty and geometric inconsistencies in practical scenarios. In this work, we address these limitations by introducing a localization framework that handles uncertainty in both pose prior and scene geometry.

Uncertainty Quantification in Radiance Fields. Recent studies [15, 17, 57, 58, 60] have proposed diverse strategies to quantify uncertainty within Radiance Fields. NeRF-based methods [22, 57, 60] estimate the reliability of scene reconstruction by modeling uncertainty along casted rays, typically through ray-wise density distributions. Few-view uncertainty estimation methods [46, 58] model epistemic uncertainty using variational inference. Bayes' Rays [15] estimates spatial uncertainty by applying a perturbation field over a spatial grid and interpolating uncertainty at query points. Similarly, FisherRF [17] computes Fisher information over NeRF and 3DGS to measure view-dependent uncertainty.

However, these uncertainty-quantification approaches are primarily designed for improving scene reconstruction, either by filtering out outliers [15, 20, 35] and distractors or by guiding active mapping [22, 60] and next-view selection [17, 40] for autonomous reconstruction, rather than addressing uncertainty for camera localization. Moreover, since most formulations depend on the query-point representation inherent to NeRF [36], they are not directly compatible with recent explicit models such as 3D Gaussian Splatting [19]. In contrast, our work investigates how geometric and pose uncertainties jointly affect localization reliability and introduces a probabilistic refinement framework that explicitly incorporates these uncertainties into the localization process.

3. Method

3.1. Overview

Pose Refinement on 3DGS. In this work, we consider the uncertainty in camera pose refinement on 3DGS and propose a probabilistic pose refinement framework, as illustrated in Fig. 2. Pose refinement utilizing Gaussian Splatting [16, 26, 30] typically rely on deterministic pipelines, where the pose estimation module predicts the pose prior $\mathbf{T}_{\text{prior}} = [\mathbf{R}_i, \mathbf{t}_i] \in SE(3)$, which is then refined through Gaussian Splatting $\mathcal{G}$. In these pipelines, a rendered image I_r from estimated pose $\mathbf{T}_{\text{prior}}$ and query image I_q are then used to extract 2D-2D correspondences $C_{q,r}$ by leveraging 2D matching modules $\mathcal{M}$ like MASt3r [25] and LoFTR [49]. These 2D-2D pixel correspondences $C_{q,r} = \{(\mathbf{q}_i, \mathbf{r}_i)\}$ are subsequently extended to 2D-3D correspondences $\mathcal{C} = \{(\mathbf{q}_i, \mathbf{X}_i)\}$ using the rendered depth map D_r and the camera intrinsics K:

$$z_i = D_r(\mathbf{r}_i), \quad \mathbf{X}_i = \mathbf{T}_{\text{prior}}\big(z_i K^{-1} \tilde{\mathbf{r}}_i\big), \tag{1}$$

where $\tilde{\mathbf{r}}_i = [u_i, v_i, 1]^\top, \mathbf{r}_i = (u_i, v_i)$ denotes the pixel in the rendered image, Finally, the pose prior $\mathbf{T}_{\text{prior}}$ is refined using a PnP-RANSAC solver [11, 12, 24] to predict the camera pose $\mathbf{T}_{\text{refined}}$.

Pose Prior Uncertainty. 3DGS-based pose refinement pipeline does not require any additional training and is both simple and efficient, which is why it is adopted in many recent works. However, this pipeline makes two key assumptions: an accurate pose prior and the precise geometry of the scene. First, if the pose prior has a significant error or is located in an area with high occlusion within the scene, matching with the query image fails, or the matching quality may degrade. As shown in Fig. 2, the refined pose generated

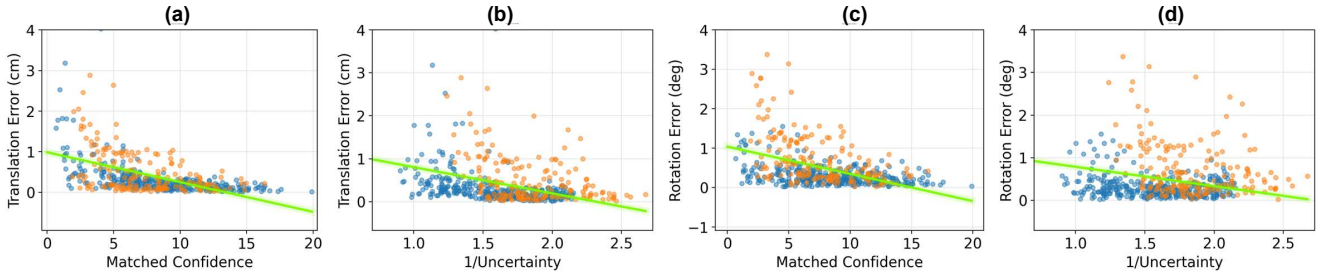

Figure 3. **Distributions of Error–Confidence and Error–Uncertainty.** We visualize the relationship between translation error and the aggregated matcher confidence (a,c) or aggregated depth uncertainty (b,d) for frames from the Kings (blue) and Hospital (orange) scenes of the Cambridge dataset. For each frame, the x-axis value represents the sum of confidence or uncertainty across all its 2D–2D correspondences. The green trend line illustrates that higher confidence and lower uncertainty consistently correlate with smaller localization errors.

from such deterministic refinement is highly sensitive to the initial pose prior from the pose estimator [5, 6, 8]. This issue arises from the reliance on a single deterministic pose estimate. To address this, we utilize modified Monte Carlo Localization [21, 52], which helps resolve such ambiguities and uncertainties by sampling multiple pose hypotheses and refining them probabilistically.

Geometric Uncertainty. The 3DGS-based pose refinement pipeline is also dependent on the geometric quality of the 3DGS representation $\mathcal{G}$. In the existing pipeline, pose prior is refined via PnP-RANSAC with the correspondence between 3D coordinate map X_r from Eq. (1). Even if the image synthesis is accurate, incorrect geometry can lead to erroneous 3D point information [15, 17, 58], which negatively impacts the final pose estimate. Building on these insights, we explore depth uncertainty in anchor-based Gaussian Splatting [33]. Through the depth uncertainty estimated across the scene, we propose an uncertainty-guided point sampling strategy for pose refinement. This sampling scheme explicitly exploits geometric reliability to mitigate failure cases caused by erroneous 3D points during PnP optimization process, as detailed in Sec. 3.2.

3.2. Monte Carlo Refinement

To account for pose prior uncertainty, we adopt an MCL-inspired strategy that generates multiple pose hypotheses from the initial pose, refines each through our optimization procedure, and evaluates them via importance sampling. MCL represents the camera pose as a set of weighted particles $\mathcal{P} = \{(\mathbf{T}^{(m)}, w^{(m)})\}_{m=1}^{M}$, where each $\mathbf{T}^{(m)} \in SE(3)$ denotes a sampled particle and $w^{(m)}$ its importance weight. However, standard MCL frameworks are computationally inefficient for 6DoF pose in this high-dimensional state space. The traditional prediction phase, which typically involves simple random perturbation or sampling from a wide prior, is non-directed and requires a prohibitively large number of particles to adequately cover the $SE(3)$ manifold and converge. To address this, we replace prediction phase with local optimization as in Sec. 3.3. This step guides each particle toward a nearby mode in the likelihood distribution. Crucially, this optimization acts as a directed pre-correction mechanism, significantly minimizing the uncertainty introduced by the wide prior.

After the pre-correction step, the importance weight $w^{(m)}$ of each particle m needs to be identified to sample the most likely particle. Rather than applying photometric error [28, 50, 59], which necessitates additional rendering and is highly susceptible to domain shift and scene appearance variations, we utilize two key metrics for efficient evaluation. Specifically, we leverage the matching confidence score S_m, which is a byproduct inherently generated by the 2D matching module $\mathcal{M}$, and the geometric uncertainty map U_m, described in Sec. 3.3. As illustrated in Fig. 3, a high matching confidence score S_m and low geometric uncertainty U_m strongly correlate with a reduced pose error, validating their use as reliable weighting factors. These metrics are then combined to quantify the fitness of the refined pose and determine whether the particle belongs to a high-likelihood region near the true camera pose.

Consequently, the importance weight $w^{(m)}$ for each refined particle m is defined as the product of the two complementary factors, normalized over the particle set $\mathcal{P}$:

$$w^{(m)} = \frac{\sum_{i=1}^{n_m} S_m(r_i) \cdot (1 - U_m(r_i))}{\sum_{j=1}^{M} \sum_{i=1}^{n_j} S_j(r_i) \cdot (1 - U_j(r_i))}, \qquad (2)$$

where r_i denotes the 2D-2D correspondence pixels on the rendered image. Following weight assignment, the final pose estimate is derived by resampling the particles based on $w^{(m)}$. The final estimated camera pose, $\hat{\mathbf{T}}_{\text{final}}$, is then typically extracted from the resulting particle set by either selecting the particle with the highest weight or by calculating the weighted mean of the particle poses in $SE(3)$ at the final iteration.

3.3. Fisher Information Guided Pose Refinement

Fisher Information Approximation. To identify geometric uncertainty from 3DGS into pose refinement, we

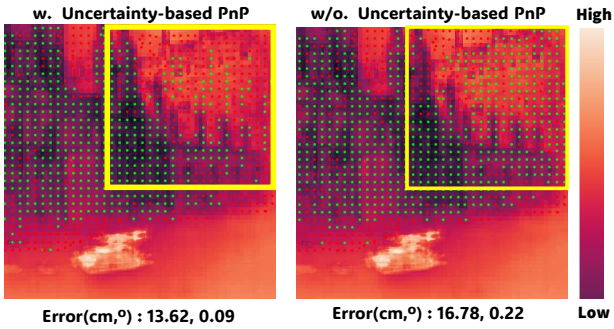

Figure 4. **Effect of Uncertainty-based PnP.** We visualize the rendered uncertainty map together with 2D–2D correspondences projected onto the synthesized view. Green and red points denote inlier and outlier points, respectively. When applying our uncertainty-guided PnP, correspondences located in high-uncertainty regions are naturally suppressed, yielding more reliable inliers and improved pose estimate results.

extend the Fisher Information–based depth uncertainty to an anchor-based Gaussian Splatting representation [33] inspired by FisherRF [17]. Let the overall 3DGS model parameters be expressed as

$$\theta = (\phi,\, g), \qquad g = D(\phi), \tag{3}$$

where ϕ denotes the anchor parameters and g the Gaussian attributes generated from decoder D. Following [17], the Fisher Information can be approximated under standard regularity conditions [44] as the expected negative Hessian of the log-likelihood:

$$\mathcal{I}(x,\theta) = -\,\mathbb{E}\big[\nabla_\theta^2 \log p(\mathbf{y} \mid \mathbf{x},\theta)\big] = \mathrm{H}''[y|x,\theta], \tag{4}$$

where x denotes the rendering pose. The Hessian matrix H'' is computed from the Jacobian of the rendered outputs with respect to θ, which can be efficiently obtained via a single backward pass for a given view. In practice, we apply a Laplace approximation [9, 34] that replaces the full Hessian with its diagonal elements and a small regularization term:

$$\mathrm{H}''[y|x,\theta] \simeq \mathrm{diag}\big((\nabla_\theta f)^\top (\nabla_\theta f)\big) + \lambda I. \tag{5}$$

Aggregating Depth Uncertainty. To quantify the geometric uncertainty, we aggregate Fisher Information $\mathcal{I}(x_i,\theta)$ from the given training views $x_i \in X$ in a global matrix G, defined as $\mathrm{G} = \sum_{i=1}^{N_{\text{views}}} \mathrm{H}''[y|x_i,\theta]$. Then, the uncertainty U of the estimated view x_q is computed using the 3DGS rendering formulation, where each Gaussian's geometric uncertainty is weighted by its depth contribution along the ray. By normalizing over the total number of Gaussians N, this operation effectively projects the accumulated 3D uncertainty into a per-pixel 2D uncertainty map:

$$U = \frac{1}{N}\sum_{i=1}^{N} \mathrm{tr}(\mathrm{G}_i)\, d_i \alpha_i \prod\nolimits_{j=1}^{i-1}(1-\alpha_j). \tag{6}$$

Here, d_i and G_n represent the depth of gaussian i and the submatrix of G corresponding to the ith Gaussian point seen from the rendering view.

Pose Estimation via Geometric Uncertainty. To incorporate the Fisher-based geometric uncertainty G into the pose optimization procedure, the 2D-3D correspondences $\mathcal{C} = \{(\mathbf{q}_i, \mathbf{X}_i)\}$ from Eq. (1) are associated with a specific geometric uncertainty $U(r_i)$. This $U(r_i)$ quantifies the local geometric reliability of the reconstructed point $\mathbf{X}_i$. Unlike Simple RANSAC-Sampler, which relies on uniform sampling, our framework introduces a probabilistic bias from uncertainty fields. We construct sampling weights s_i for each correspondence $\mathcal{C}$ based on this geometric reliability:

$$s_i = e^{-\beta\, \bar{U}(r_i)} + \epsilon, \tag{7}$$

where $\bar{U} \in [0,1]$ denotes the normalized uncertainty with all the uncertainty values of the matched pixels. This formulation biases the sampling stage towards geometrically reliable correspondences by assigning higher weights s_i. At each refinement iteration, the minimal 2D-3D subset $\mathcal{S}_k \subset \mathcal{C}$ is sampled according to the probabilistic weights s_i. The estimated pose $\mathbf{T}_k$ is then calculated using the EPnP algorithm [24] iteratively. This iterative process ensures that each newly generated hypothesis is inherently guided toward stable geometric anchors, as illustrated in Fig. 4. Furthermore, we extend this bias to the consensus evaluation. The final pose $\mathbf{T}_{\text{refined}}$ is selected by maximizing the weighted consensus $\sum s_i$.

4. Experiments

In this section, we present the localization performance and analysis of our proposed pose refinement pipeline. Sec. 4.1 reports localization accuracy across both indoor and outdoor benchmarks. Sec. 4.2 provides a detailed ablation study, examining the key components of our framework, including the comparison against simple iterative refinement, the influence of particle count, and the impact of different matching modules.

Datasets. We evaluate our method on three widely used visual localization benchmarks: 7Scenes [47], 12Scenes [54], and Cambridge Landmarks [18]. The 7Scenes dataset [47] consists of seven indoor environments, from offices to living rooms, with spatial volumes ranging from approximately 1 m³ to 18 m³. The 12Scenes [54] dataset extends this setting to larger indoor environments (14 m³ – 79 m³), providing more complex scene layouts and viewpoint diversity. The Cambridge Landmarks [18] dataset contains four large-scale outdoor scenes with spatial extents from 875 m² to 5600 m², capturing various urban landmarks under significant illumination and viewpoint changes.

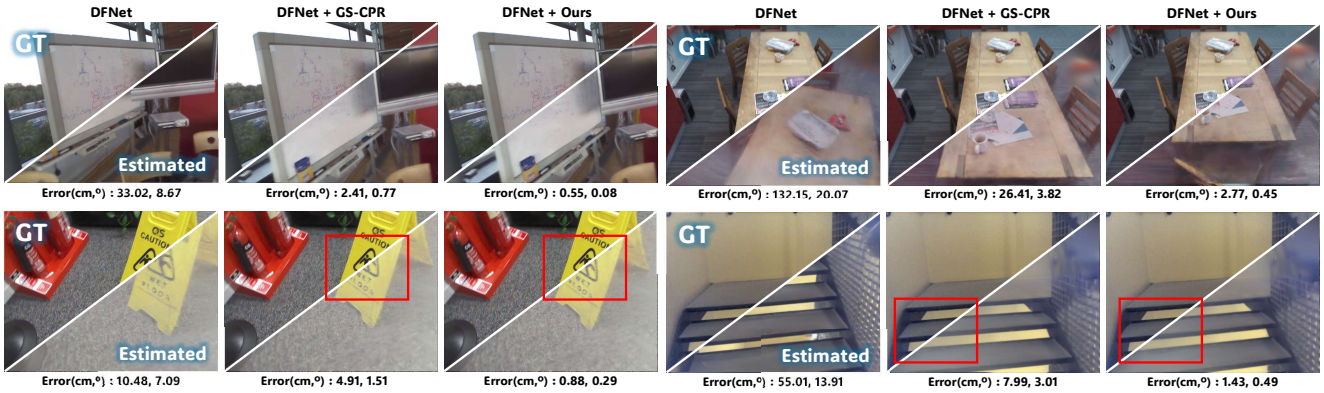

Figure 5. **Visualization of Localization Quality on the 7-Scenes dataset.** Each triplet of images compares the ground-truth view (top-left) with the view rendered from (i) the input pose prior, (ii) the baseline GS-CPR refinement, and (iii) our UGS-Loc refinement (bottom-right). A closer alignment along the diagonal boundary indicates better pose accuracy.

Table 1. **Localization Results on 7Scenes dataset**. We report the median translation errors (cm) and rotation errors (°). The best and second-best results are in **bold** and underline.

	Methods	Chess	Fire	Heads	Office	Pumpkin	Redkitchen	Stairs	Avg. ↓ [cm/°]
APR	PoseNet [18]	10/4.02	27/10.0	18/13.0	17/5.97	19/4.67	22/5.91	35/10.5	21/7.74
	DFNet [6]	3/1.12	6/2.30	4/2.29	6/1.54	7/1.92	7/1.74	12/2.63	6/1.93
	Marepo [8]	1.9/0.83	2.3/0.92	2.1/1.24	2.9/0.93	2.5/0.88	2.9/0.98	5.9/1.48	2.9/1.04
	RAP [26]	1/0.84	6/3.44	4/5.48	5/1.92	4/1.71	7/2.11	9/2.10	5/2.51
SCR	DSAC* [4]	0.5/0.17	0.8/0.28	0.5/0.34	1.2/0.34	1.2/0.28	0.7/0.21	2.7/0.78	1.1/0.34
	ACE [5]	0.5/0.18	0.8/0.33	0.5/0.33	1.0/0.29	1.0/0.22	0.8/0.2	2.9/0.81	1.1/0.34
	GLACE [55]	0.6/0.18	0.9/0.34	0.6/0.34	1.1/0.29	0.9/0.23	0.8/0.20	3.2/0.93	1.2/0.36
NRP	FQN-MN [13]	4.1/1.31	10.5/2.97	9.2/2.45	3.6/2.36	4.6/1.76	16.1/4.42	139.5/34.67	28/7.3
	CrossFire [38]	1/0.4	5/1.9	3/2.3	5/1.6	3/0.8	2/0.8	12/1.9	4.4/1.38
	pNeRFLoc [61]	2/0.8	2/0.88	1/0.83	3/1.05	6/1.51	5/1.54	32/5.73	7.3/1.76
	DFNet + $NeFeS_{50}$ [7]	2/0.57	2/0.74	2/1.28	2/0.56	2/0.55	2/0.57	5/1.28	2.4/0.79
	NeRFMatch [63]	0.9/0.3	1.1/0.4	1.5/1.0	3.0/0.8	2.2/0.6	1.0/0.3	10.1/1.7	2.8/0.7
	MCLoc [53]	2/0.8	3/1.4	3/1.3	4/1.3	5/1.6	6/1.6	6/2.0	4.1/1.43
	DFNet + GS-CPR [30]	0.7/0.20	0.9/0.32	0.6/0.36	1.2/0.32	1.3/0.31	0.9/0.25	2.2/0.61	1.1/0.34
	Marepo + GS-CPR [30]	0.6/0.18	0.7/0.28	0.5/0.32	1.1/0.29	1.0/0.26	0.8/0.21	1.5/0.44	0.9/0.28
	ACE + GS-CPR [30]	0.5/0.15	0.6/0.25	0.4/0.28	0.9/0.26	1.0/ 0.23	0.7/0.17	1.4/0.42	0.8/0.25
	STDLoc [16]	0.46/0.15	0.57/0.24	0.45/0.26	0.86/0.24	0.93/0.21	0.63/0.19	1.42/0.41	0.76/0.24
	DFNet + **Ours**	**0.36**/0.16	0.48/**0.20**	**0.37/0.25**	**0.77/0.21**	0.81/**0.18**	0.59/**0.15**	1.21/0.35	0.66/**0.21**
	Marepo + **Ours**	0.37/**0.12**	0.48/**0.20**	**0.36**/0.26	0.78/0.22	0.81/**0.18**	0.59/**0.15**	1.13/0.34	0.65//**0.21**
	ACE + **Ours**	0.37/**0.12**	**0.47/0.20**	**0.36/0.25**	**0.77**/0.22	**0.79/0.18**	**0.58/0.15**	**1.11/0.33**	**0.64/0.21**

Implementation Details. In Monte Carlo Refinement, although lightweight matching [10, 29, 42] modules are employed, increasing the number of iterations $\mathcal{N}$ or particles m leads to a substantial rise in inference time, due to the repeated matching and pose–optimization steps. Therefore, we maintain the entire pose refinement process to $\mathcal{N} = 2$ with $m = 8$ particles throughout all experiments. To further improve efficiency, the pose refinement and weighting of each particle are processed in parallel using multiprocessing as [53]. For the benchmark experiments, we use MASt3r [25] as the matching module without finetuning, while additional studies on alternative matching modules are presented in Sec. 4.2 and supplementary material. For the perturbation noise of Monte Carlo Refinement, the first iteration samples translation and rotation perturbations from a uniform distribution with ranges of 10cm and 0.01°, respectively. while subsequent iterations use 1 cm translation and 0.01° rotation noise. To derive the Fisher Information $\mathcal{I}(\theta)$ on Scaffold-GS parameters [33], only the anchor feature and the local Gaussian offset is explored as the parameters θ. All experiments are conducted on an NVIDIA RTX A5000 GPU. Following GS-CPR [30], we train the Scaffold-GS [33] model for 30k iterations and adopt the exposure adaptation strategy. With our standard localization configuration ($m = 8$), rendering all particle views requires 190ms, and the batchified MASt3r inference

Table 2. **Average Recall on 7-Scenes.** We report the recall rate within the $[2\text{cm}, 2°]$ and $[5\text{cm}, 5°]$ error thresholds across the 7Scenes.

Datasets	7scenes	
Methods	Avg. Acc ↑ $[5\text{cm}, 5°]$	Avg. Acc ↑ $[2\text{cm}, 2°]$
DFNet	43.1	8.4
DFNet + GS-CPR	94.2	76.5
DFNet + GS-CPR^2	98.4	86.1
DFNet + **UGS-Loc**	**99.7**	**95.1**
ACE	97.1	83.3
ACE + GS-CPR	**100**	93.1
ACE + GS-CPR^2	**100**	93.1
ACE + **UGS-Loc**	**100**	**95.6**

takes 492ms. The uncertainty-guided PnP optimization with MASt3r matcher, executed in a multi-processing setting, adds 377ms. Overall, the end-to-end inference time is 1.1s per iteration. The total runtime can be adjusted by varying the number of particles. Notably, although MCLoc adopts a similar Monte Carlo–based refinement pipeline, its 80 iterations procedure requires around 2.4s per query, while our method remains within a practical runtime range under comparable settings.

Baselines. We compare its localization performance with state-of-the-art approaches across three benchmark datasets [18, 27, 47]. The comparison includes absolute pose regression (APR) method [45], including PoseNet [18], DFNet [6], LENS [37], and RAP [26]; scene coordinate regression (SCR) methods such as DSAC* [4], ACE [5], and GLACE [55]; and neural rendering-based post-refinement (NRP) methods [7, 13, 16, 30, 38, 53, 61, 63], which leverage 2D-3D correspondence derived from neural scene representation or refine the initial pose by synthesising a novel view.

4.1. Benchmark Results

7-scenes [14, 47]. As shown in Tab. 1, we present the median translation and rotation errors for each scene. Our proposed method, UGS-Loc, achieves approximately 40% improvement from the prior poses estimated from ACE [5], DFNet [6], and merapo [8] in terms of both median translation and rotation errors. Furthermore, when compared to GS-CPR [30], a purely deterministic radiance field–based pose refinement framework, our method shows an additional improvement about 20% for three initial pose estimators [5, 6, 8], despite employing a similar refinement procedure. Also, we demonstrate average accuracy of a $2\text{cm}, 2°$ and $5\text{cm}, 5°$ psoe error threshold in Tab. 2. In Tab. 2, we additionally include the results of GS-CPR^2, which applies the refinement structure of GS-CPR in an iterative manner for a fair comparison. GS-CPR^2 performs relocalization iteratively by using the refined pose as the new initial pose. Our proposed UGS-Loc demonstrates higher accuracy than the original non-iterative refinement structure and also outperforms the iterative GS-CPR^2 by 10% under the $2\text{cm}, 2°$ threshold. This improvement stems not from iterative refinement but from effectively addressing the limitations of the deterministic framework through our proposed probabilistic formulation.

Table 3. **Localization Results on 12Scenes dataset.** We present the average percentage of frames whose estimated poses satisfy the $[2\text{cm}, 2°]$ and $[5\text{cm}, 5°]$ thresholds, along with the median translation and rotation errors (cm/°) across the 12Scenes.

Methods	Avg. Err ↓ [cm/°]	Avg. ↑ $[5\text{cm}, 5°]$	Avg. ↑ $[2\text{cm}, 2°]$
Marepo	2.1/1.04	95	50.4
DSAC*	0.5/0.25	99.8	96.7
ACE	0.7/0.26	**100**	97.2
GLACE	0.7/0.25	**100**	97.5
Marepo + GS-CPR	0.7/0.28	98.9	90.9
Marepo + **UGS-Loc**	0.5/0.21	99.4	95.3
ACE + GS-CPR	0.5/0.21	**100**	98.7
ACE + **UGS-Loc**	**0.4/0.18**	**100**	**99.1**

Table 4. **Localization Results on Cambridge Landmark dataset.** We report the median translation errors (cm) and rotation errors (°). The best results are in **bold** and second-best results are indicated with an underline.

	Methods	Kings	Hospital	Shop	Church	Avg. ↓ [cm/°]
APR	PoseNet [18]	93/2.73	224/7.88	147/6.62	237/5.94	175/5.79
	MS-Transformer [45]	85/1.45	175/2.43	88/3.20	166/4.12	129/2.80
	LENS [37]	33/0.5	44/0.9	27/1.6	53/1.6	39/1.15
	DFNet [6]	73/2.37	200/2.98	67/2.21	137/4.02	119/2.90
	RAP [26]	58/0.87	81/1.20	33/1.64	60/1.83	58/1.39
SCR	ACE [5]	29/0.38	31/0.61	5/0.3	19/0.6	21/0.47
	GLACE [55]	20/0.32	20/0.41	5/0.22	9/0.3	14/0.32
NRP	FQN-MN [13]	28/0.4	54/0.8	13/0.6	58/2	38/1
	CrossFire [38]	47/0.7	43/0.7	20/1.2	39/1.4	37/1
	DFNet + NeFeS$_{30}$ [7]	37/0.64	98/1.61	17/0.60	42/1.38	49/1.06
	DFNet + NeFeS$_{50}$ [7]	37/0.54	52/0.88	15/0.53	37/1.14	35/0.77
	HR-APR [31]	36/0.58	53/0.89	13/0.51	38/1.16	35/0.78
	MCLoc [53]	31/0.42	39/0.73	12/0.45	26/0.8	27/0.6
	RAP$_{ref}$ [26]	18/0.38	22/0.42	8/0.39	16/0.47	16/0.42
	DFNet + GS-CPR [30]	23/0.32	42/0.74	10/0.36	27/0.62	26/0.51
	ACE + GS-CPR [30]	20/0.29	21/0.40	5/0.24	13/0.40	15/0.33
	DFNet + **Ours**	18.7/0.19	14.5/**0.29**	**3.9**/0.15	**5.5/0.17**	10.7/**0.20**
	ACE + **Ours**	**17.8/0.18**	**13.8**/0.30	4.2/**0.16**	6.3/0.20	**10.5**/0.21

12-scenes [54]. As shown in Tab. 3, we report the quantitative localization performance using Merapo [8] and ACE [5] as pose priors. Given the strong baseline accuracy of SCR-based methods [3, 55] on the $[5\text{cm}, 5°]$ threshold, we focus on the strict threshold $[2\text{cm}, 2°]$, where our approach achieves the highest accuracy of 99.1%. Moreover, our method attains the lowest median translation and rotation errors across all scenes, outperforming the conventional refinement pipeline GS-CPR by delivering more precise and stable localization results.

Cambridge Landmark [18]. As shown in Tab. 4, we evaluate localization performance using DFNet [6] and ACE [5] as pose priors. UGS-Loc reduces the median translation error of the baseline GS-CPR [30] by approximately 30%,

Table 5. **Evaluating the Number of Particles for Refinement.** We report the median translation and rotation errors (cm/°) on the Cambridge Landmarks dataset across different particle counts used in the Monte Carlo Refinement. PP and NP denote pose prior and the number of particles

PP	NP	Kings	Hospital	Shop	Church	Avg. ↓ [cm/°]
DFNet	2	18.7/0.20	16.24/0.36	5.5/0.20	6.6/0.20	11.8/0.24
	4	19.1/0.19	15.1/0.29	4.6/0.19	6.0/0.18	11.2/0.21
	8	18.7/**0.19**	14.50.29	**3.9/0.15**	5.5/0.17	10.7/**0.20**
	16	**18.2**/0.20	**13.8/0.27**	4.2/0.16	**5.1/0.16**	**10.3/0.20**
ACE	2	17.8/0.21	14.8/0.30	4.0/0.18	6.9/0.23	10.9/0.51
	4	18.8/0.19	14.3/0.28	3.7/0.16	6.8/0.21	10.7/0.21
	8	17.8/**0.18**	13.8/0.30	4.2/0.16	6.3/0.20	10.5/0.21
	16	**17.0/0.18**	**11.9/0.28**	**3.4/0.15**	**6.1/0.19**	**9.6/0.20**

demonstrating strong refinement capability. Notably, our framework converges reliably within only two iterations, consistently achieving high accuracy regardless of the quality of the initial pose prior. Although DFNet's localization performance is significantly weaker than ACE, often differing by five times, our refinement module delivers similarly accurate results for both priors. This illustrates the ability of UGS-Loc to mitigate the uncertainty introduced by suboptimal pose priors. Remarkably, even in challenging cases such as the Shop scene, UGS-Loc with DFNet prior surpasses the refinement results obtained using ACE, further validating the robustness of our approach to prior noise.

4.2. Ablation Study

Comparison with Simple Iterative Refinement. We investigate whether the performance gain of our UGS-Loc simply stems from multiple refinement iterations. To this end, we conduct an experiment on the Cambridge Landmarks dataset by iteratively applying the deterministic baseline pipeline [30] several times and comparing it with our proposed probabilistic refinement. As shown in Fig. 6, even though iterative refinement with GS-CPR gradually reduces translation and rotation errors, it quickly saturates after the first iteration. In contrast, our method achieves both consistent convergence and lower final errors, demonstrating that the improvement originates from our framework rather than from repeated deterministic optimization.

Effect of Particle Count on Pose Refinement. We analyze the effect of the number of particles used in the Monte Carlo Refinement. As shown in Tab. 5, we vary the number of particles from 2 to 16 while keeping the number of refinement iterations as 2. We observe that increasing the number of particles consistently improves both translation and rotation accuracy across all scenes. In particular, the performance gain is most pronounced when increasing the particle count. This suggests that using multiple pose hypotheses helps better capture the uncertainty of the pose prior and avoids convergence to suboptimal local minima.

Table 6. **Evaluation with Different 2D Matching Module.** We compare our probabilistic refinement framework using sparse feature matchers (SuperPoint (SP) [10] + LightGlue (LG) [29]) and dense correspondence model (MASt3r [25]).

Prior	Matchers	Kings	Hospital	Shop	Church	Avg. ↓ [cm/°]
Baseline		23/0.32	42/0.74	10/0.36	27/0.62	26/0.51
DFNet	SP + LG	20/0.22	17/0.31	5.5/0.26	7.7/0.25	13/0.26
DFNet	MASt3r	19/**0.19**	15/0.32	4.3/0.17	**5.8/0.18**	11/0.22
ACE	SP + LG	19/0.23	15/**0.27**	**4.0**/0.19	7.8/0.26	11/0.26
ACE	MASt3r	**18**/0.20	**14** /0.30	4.5/**0.16**	6.5/0.21	11/0.22

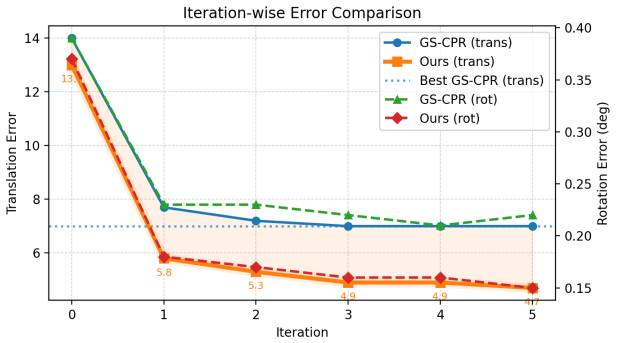

Figure 6. **Comparison with Iteration-wise Pose Refinement.** We visualize median translation and rotation errors over iterations on Church scene of the Cambridge Landmarks, comparing our UGS-Loc with the iterative refinement strategy adopted in GS-CPR [30].

Various Matching Modules. As shown in Tab. 6, our method demonstrates localization results compared to a lightweight sparse feature matcher (SuperPoint [10] + LightGlue [29]). While MASt3r tends to yield slightly higher accuracy due to its strong context reasoning and dense feature aggregation, our uncertainty-aware refinement achieves comparable localization accuracy even with lightweight sparse matchers. We report GS-CPR with MASt3r as baseline.

5. Conclusion

In this paper, we investigated the overlooked impact of prior pose and geometry uncertainty in 3D Gaussian Splatting-based pose refinement. To alleviate these issues, we proposed a probabilistic relocalization framework that combines Monte Carlo sampling with Fisher Information–guided PnP optimization. This design handles uncertainty without requiring any retraining or scene-specific adjustments. Extensive experiments across indoor and outdoor benchmarks demonstrate that our method consistently improves localization accuracy and remains robust regardless of the choice of pose prior or matching module. Overall, our findings highlight that incorporating uncertainty into the 3DGS pipeline yields a more stable and reliable pose refinement process.

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